

Prompt Library for Content Generation

Mini Project – Generative AI

Name: Kritika Gahlawat

Introduction

This is my prompt library for common content generation tasks. I have used different prompt engineering techniques such as role prompting, few-shot prompting, RTCF and constraints. I have also used placeholders so that the prompts can be reused.

1. LinkedIn Post for a Student Project

Use Case: To write a LinkedIn post about a college project.

Techniques: Role Prompting + Few-Shot + Constraints

Template:

You are a LinkedIn content writer. Write a LinkedIn post about my project Library Management System, made using Java and MySQL. The audience is college students and recruiters. Mention what I learned: database connectivity, CRUD operations and problem-solving. Keep it around 100–120 words. Use a simple and positive student tone. Start with a short opening and end with a question.

Sample Output:

Project Highlight: Library Management System 📖💻

Excited to share one of my projects — a **Library Management System** developed using **Java and MySQL**!

Working on this project gave me practical experience in connecting a Java application with a MySQL database and managing data efficiently. I learned how to implement **CRUD operations** (Create, Read, Update, and Delete) and improved my **problem-solving skills** by handling different challenges during development.

This project helped me understand how programming concepts can be applied to build a useful real-world application. It was a great learning experience and strengthened my confidence in Java and database development.

What project helped you learn the most during college? ✍️

Iteration: V1 → V2: In V1 I only asked for a LinkedIn post. In V2 I added the audience, tone, word limit and an example. This made the result more focused.

2. Instagram Post for a College Event

Use Case: To create a short Instagram caption for a college event.

Techniques: Role Prompting + Constraints

Template:

Act as a social media manager for a college. Write an Instagram caption for an AI Awareness Workshop on 20 August at Seminar Hall 2. The audience is college students. The purpose is to help students understand the practical uses of Generative AI. Keep it between 60 and 80 words. Add 2 emojis, 5 hashtags and a short call to action. Keep the language simple.

Sample Output:

 **AI Awareness Workshop – Learn, Explore & Create!**

Curious about how Generative AI can help you in college and beyond? Join our **AI Awareness Workshop** on **20 August** at **Seminar Hall 2**. Discover practical uses of AI for learning, creativity, research, productivity, and everyday tasks. Come with your questions and explore how to use AI effectively and responsibly. 💡

Save the date and join us!

#AIWorkshop #GenerativeAI #StudentLife #AIForStudents #CollegeEvent

Iteration: The constraints make the caption short and suitable for Instagram.

3. Blog Introduction

Use Case: To write the first paragraph of a student blog.

Techniques: Role Prompting + RTCF + Constraints

Template:

You are a blog writer. Write an introduction for a blog about How Beginners Can Start Learning Python. The readers are college students. The purpose of the blog is to help beginners understand how to start learning Python. Start with a question or common problem. Explain why the topic is useful and briefly say what the blog will cover. Write 120–150 words in simple language.

Sample Output:

Have you ever wanted to learn Python but felt confused about where to start? With so many tutorials, courses, and programming resources available online, choosing the right path can be difficult, especially for college students who are new to coding.

Python is one of the most beginner-friendly programming languages and is widely used in areas such as web development, data science, artificial intelligence, and automation. Learning Python can help students build programming skills, work on projects, and prepare for future academic and career opportunities.

In this blog, we will explore a simple way to start learning Python from the basics. We will cover how to choose learning resources, practice coding regularly, understand important concepts, build small projects, and stay motivated while learning.

Iteration: V1 → V2: I added the target readers, purpose, structure and word limit. The second version gave a more useful introduction.

4. Cold Outreach Email

Use Case: To contact a company or professional for an opportunity or advice.

Techniques: Role Prompting + Constraints + RTCF

Template:

You are a career mentor. Write a short email to Ms. Sharma, who is an HR Manager at ABC Technologies. I am a final-year M.Tech student and my skills are Python, SQL and Machine Learning. I am contacting her because I am interested in internship opportunities. Mention my project Machine Learning Project. I want to ask about internship opportunities and advice for students. Keep it under 120 words. Be polite and confident, not desperate.

Sample Output:

Dear Ms. Sharma,

I hope you are doing well. I am a final-year M.Tech student with skills in Python, SQL, and Machine Learning. I am interested in exploring internship opportunities at ABC Technologies and would be grateful to learn about any suitable openings.

I have also worked on a **Machine Learning Project**, which helped me develop practical experience and strengthen my technical skills.

If possible, I would appreciate any advice you could share for students looking to begin their careers in this field.

Thank you for your time. I look forward to hearing from you.

Best regards,
Kritika

Iteration: V1 → V2: The first prompt was just “write a cold email for a job”. I added my background, reason, project, request and word limit, so the email sounded more natural.

5. Follow-up Email

Use Case: To politely follow up after an application or previous email.

Techniques: Role Prompting + Constraints

Template:

You are a professional communication assistant. Write a follow-up email to Ms. Sharma about my internship application. The previous interaction was last week. I am following up because I have not received an update yet. Keep it below 90 words. Be polite and do not sound demanding. Ask for an update.

Sample Output:

Dear Ms. Sharma,

I hope you are doing well. I am writing to politely follow up on my internship application, which I shared last week. I understand you may be busy, but I wanted to check if there has been any update regarding the application.

I remain interested in the opportunity and would appreciate any information you can share when convenient.

Thank you for your time and consideration.

Best regards,
Kritika

Iteration: The word limit and tone instruction help keep the email short and polite.

6. Product Description

Use Case: To write a simple description for a product.

Techniques: Role Prompting + Few-Shot + Constraints

Template:

Act as an e-commerce copywriter. Write a description for a Study Planner Notebook. Details: a notebook with weekly planning pages, daily task space and note pages. Target customer: college students. Main benefit: helping students organise their study work. Use this style: "A simple everyday product made for people who want something useful and easy to use." Write 80–100 words. Include 3 feature points. Do not make up any product details.

Sample Output:

A simple study planner made for college students who want something useful and easy to use. The Study Planner Notebook helps you organise your study work by giving you dedicated spaces to plan, manage tasks, and write important notes. It is designed to keep your academic work in one place and make planning easier without adding complexity.

Features:

- Weekly planning pages for organising your study schedule
- Daily task space for keeping track of study work
- Note pages for writing down important information

A practical choice for students who want a simple way to stay organised.

Iteration: The example and constraints make the product description follow a simple and consistent style.

7. YouTube Title and Description

Use Case: To create titles and a description for a YouTube video.

Techniques: Role Prompting + Few-Shot + Constraints

Template:

You are a YouTube content creator for students. Create 5 title options and one description for a video about How to Start Learning Python. The audience is college students and beginners. The main takeaway is that regular practice and learning the basics can make Python easier. Use this title style: "3 Simple Ways to Learn Python Faster". Keep each title under 60 characters. Do not use clickbait. Keep the description between 80 and 120 words.

Sample Output:

Title Options

1. 3 Simple Ways to Start Learning Python
2. 5 Easy Steps to Learn Python for Beginners
3. 4 Simple Tips to Learn Python from Scratch
4. 3 Practical Ways to Learn Python as a Student
5. 5 Simple Habits for Learning Python Better

Description

Want to start learning Python but don't know where to begin? This video is designed for college students and beginners who want a simple and practical approach to learning Python. We will discuss why starting with the basics is important, how regular practice can improve your coding skills, and how to build confidence step by step. You will also learn how to practise consistently without feeling overwhelmed. Whether you are completely new to programming or looking to strengthen your foundation, this video will help you create a simple learning routine and make steady progress with Python.

Iteration: The first idea was only to ask for a YouTube title. I improved it by asking for 5 titles, a description and a length limit.

8. Newsletter Section

Use Case: To write a small section for a college newsletter.

Techniques: Role Prompting + Constraints

Template:

Act as a newsletter editor. Write one section about Generative AI for college students. The purpose is to explain a few simple ways students can use AI for learning and content creation. Use a friendly and informative tone. Give a heading, two short paragraphs, three bullet points and one call to action. Keep it below 180 words.

Sample Output:

Generative AI: A Helpful Tool for Students

Generative AI can be a useful learning assistant for college students when used thoughtfully. It can help explain difficult concepts in simpler language, provide examples, and support students while they revise or explore new topics.

Students can also use AI for content creation, such as developing ideas, improving drafts, or organising information. The goal is not to replace your own thinking, but to use AI as a tool that supports learning and creativity.

- **Learn better:** Ask AI to explain complex topics step by step.
- **Create ideas:** Use it to brainstorm topics for projects, presentations, or posts.
- **Improve content:** Get help with structure, clarity, and language in your drafts.

Call to action: Try one simple AI tool today and see how it can support your learning!

Iteration: The format instructions make the newsletter easy to read instead of giving one long paragraph.

9. Article Summarization

Use Case: To summarize an article for quick study.

Techniques: Role Prompting + Constraints + Iterative Refinement

Template:

You are a study assistant for college students. Summarize the article about Prompt Engineering for a college student who is learning Generative AI. Give a 2-sentence overview, 5 key points, 3 important terms with simple meanings and one practical takeaway. Keep it below 250 words. Do not add information that is not in the article.

Article:

Prompt Engineering is the skill of writing instructions clearly so that an AI model gives a focused and useful answer. A vague prompt gives a generic answer, while a specific prompt gives a more focused result.

RTCF stands for Role, Task, Context and Format. Role tells the AI who it should act as, Task tells it what to do, Context explains the situation or audience, and Format tells it how the answer should look.

The six techniques discussed are Zero-Shot Prompting, Few-Shot Prompting, Role Prompting, Chain-of-Thought, Constraints and Format Control, and Iterative Refinement. Zero-shot is useful for simple tasks. Few-shot uses examples to guide the output. Role prompting gives the AI a specific expert role. Chain-of-Thought can help with multi-step tasks. Constraints control things such as length and format. Iterative refinement means improving the same prompt instead of starting again.

Sample Output:

2-Sentence Overview

Prompt engineering is the skill of writing clear instructions so an AI model produces focused and useful answers. Using techniques such as RTCF and iterative refinement can help improve the quality and usefulness of AI outputs.

5 Key Points

1. Specific prompts generally produce more focused results than vague prompts.
2. RTCF means **Role, Task, Context, and Format**.
3. Zero-Shot Prompting is useful for simple tasks, while Few-Shot Prompting uses examples to guide output.
4. Role Prompting gives the AI a specific expert role, and Chain-of-Thought can help with multi-step tasks.

5. Constraints control aspects such as length and format, while Iterative Refinement means improving the same prompt.

3 Important Terms

- **Prompt Engineering:** Writing clear instructions for AI.
- **Few-Shot Prompting:** Using examples to guide AI output.
- **Iterative Refinement:** Improving the same prompt instead of starting again.

Practical Takeaway

Write prompts with a clear **Role, Task, Context, and Format**, then refine the prompt if the first result is not useful.

Iteration: V1: "Summarize this article." V2: I added exact sections, a word limit and a rule not to add new information. This made the summary more useful for studying.

10. Tone Rewriter

Use Case: To change a message from formal to casual or casual to formal.

Techniques: Role Prompting + Zero-Shot + Constraints

Template:

You are a communication editor. Rewrite the following formal message into a friendly and slightly casual message for a college student. Keep the meaning, facts and name the same. Do not add new information. Keep it between 50 and 70 words. Give only the rewritten text.

Text:

Dear Rahul,

I am writing to inform you that I will not be able to attend the meeting scheduled for Friday due to another academic commitment. I would like to request that the meeting be rescheduled to another convenient time.

I apologise for the inconvenience and appreciate your understanding.

Sample Output:

Hi Rahul,

I won't be able to attend the meeting scheduled for Friday because I have another academic commitment. Could we please reschedule the meeting to another convenient time? Sorry for the inconvenience, and I really appreciate your understanding.

Thanks!

Iteration: The prompt focuses on changing only the tone while keeping the original meaning unchanged.